

TORP, Norway

WMO: 014830

Lat: 59.184N

Lon: 10.255E

Elev: 289

StdP: 14.54

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.7	10.5	1.8	6.1	9.3	5.4	7.3	13.7	27.6	38.4	25.3	39.0	6.6	0	0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.3	78.6	62.4	74.9	61.6	71.4	60.2	65.6	73.2	63.9	70.6	62.6	68.3	7.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	87.1	66.2	62.3	84.7	65.5	60.7	80.0	64.2	30.6	74.0	29.4	70.9	28.3	67.2	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	20.5	18.0	4.2	81.6	4.7	5.0	0.8	85.2	-2.0	88.1	-4.6	90.9	-8.0	94.5
			3.7	68.0	4.7	2.0	0.3	69.5	-2.4	70.6	-5.0	71.8	-8.4	73.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.7	27.8	28.8	35.4	43.3	52.1	58.3	62.7	59.9	54.4	44.8	36.9	30.7
	DBStd	13.82	8.45	8.34	7.30	6.01	6.79	4.84	4.63	3.35	4.47	5.78	7.02	9.53
	HDD50	3192	687	592	456	218	54	1	0	0	14	180	393	597
	HDD65	7462	1152	1012	918	652	404	212	103	161	319	625	842	1062
	CDD50	1255	0	0	4	15	119	249	394	308	145	20	1	0
	CDD65	49	0	0	0	0	3	10	32	4	0	0	0	0
	CDH74	374	0	0	0	0	30	87	246	11	0	0	0	0
	CDH80	55	0	0	0	0	1	8	46	0	0	0	0	0
Wind	WSAvg	8.3	8.6	8.5	8.5	8.1	8.2	8.1	7.5	7.5	8.0	8.6	9.0	8.8
Precipitation	PrecAvg	38.3	3.3	2.4	1.9	2.1	2.5	2.7	3.1	4.6	3.6	4.4	4.4	3.3
	PrecMax	53.7	7.4	7.3	3.9	4.5	6.0	5.2	6.9	8.9	11.1	12.3	8.3	7.2
	PrecMin	28.6	0.6	0.3	0.0	0.6	0.1	0.2	0.5	1.3	1.5	0.3	0.7	1.1
	PrecStd	7.0	1.8	1.7	1.2	1.1	1.5	1.2	1.4	1.9	2.4	2.8	1.9	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.0	50.0	61.1	66.2	77.2	80.5	84.5	75.5	68.3	60.4	53.6	50.0
		MCWB	42.3	42.8	49.2	52.6	59.9	60.7	65.0	62.9	59.0	54.2	50.9	49.0
	2%	DB	43.2	44.5	53.6	61.9	73.2	76.8	80.6	71.9	66.2	57.4	50.3	46.8
		MCWB	41.2	38.5	44.1	49.5	58.4	60.6	64.7	61.5	57.9	53.4	49.0	45.7
	5%	DB	41.2	41.2	48.3	57.5	68.2	71.6	76.6	69.6	64.1	55.6	48.4	44.8
		MCWB	39.1	38.4	41.0	47.8	56.1	58.3	63.4	60.7	57.0	52.4	47.2	43.2
	10%	DB	39.2	39.3	44.7	53.9	64.4	67.9	73.0	67.6	62.2	53.8	46.5	42.8
		MCWB	37.4	37.6	38.7	45.7	54.5	56.2	62.2	59.6	56.6	50.8	45.3	41.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.8	43.7	50.1	53.7	63.7	64.8	68.0	66.5	64.3	57.2	52.1	48.8
		MCDB	44.2	47.6	61.6	60.5	73.5	72.6	76.4	70.0	65.2	58.5	52.6	48.9
	2%	WB	41.6	40.7	45.1	51.0	61.0	62.7	66.4	64.3	61.1	55.4	49.4	46.3
		MCDB	42.6	42.0	51.2	59.1	68.3	72.2	75.7	68.7	62.9	56.5	50.0	46.9
	5%	WB	39.7	39.1	42.2	49.1	58.7	60.8	65.0	62.8	59.4	53.6	47.4	43.3
		MCDB	40.6	40.5	45.5	56.6	64.8	67.3	73.4	66.4	61.6	55.1	48.1	44.1
	10%	WB	37.4	37.5	40.4	46.7	55.9	58.9	63.5	61.3	57.8	51.7	45.2	40.9
		MCDB	38.7	38.8	43.1	52.4	62.1	64.7	70.6	64.8	60.5	53.8	46.2	41.9
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.5	11.6	13.8	14.8	14.2	14.3	13.3	12.5	10.3	8.0	8.5
		MCDBR	9.9	11.5	17.0	18.6	20.6	19.9	20.8	16.1	15.1	11.1	10.7	8.8
		MCWBR	9.3	8.1	10.6	10.3	10.3	9.5	8.8	8.5	9.7	8.9	10.1	9.0
	5% WB	MCDBR	8.8	8.8	14.2	17.5	17.1	15.9	16.9	12.5	11.9	9.5	9.9	8.8
		MCWBR	8.9	6.8	9.8	10.2	9.5	9.0	8.6	8.2	9.5	9.1	10.1	9.3
Clear-Sky Solar Irradiance	taub	0.278	0.314	0.334	0.374	0.367	0.354	0.376	0.372	0.343	0.330	0.295	0.253	
	taud	2.238	2.318	2.365	2.324	2.419	2.477	2.435	2.442	2.512	2.445	2.296	2.159	
	Ebn at Noon	169	223	255	263	274	279	269	262	252	216	161	133	
	Edh at Noon	14	21	27	34	33	31	32	30	24	19	13	10	
All-Sky Solar Radiation	RadAvg	101	289	765	1230	1615	1844	1651	1297	858	385	134	68	
	RadStd	19	55	76	170	165	125	193	89	98	68	33	7	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page